

§ 1 绪论

§ 1.4 不确定关系

$$\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2}$$

$$\Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2}$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

例: 粒子由 Gaussian 波包 $\psi(x) = A e^{ik_0 x} e^{-\frac{x^2}{2\Delta^2}}$ 描述, 计算 $\Delta x, \Delta p$

$$\begin{aligned}\langle x \rangle &= \int_{-\infty}^{\infty} \psi^*(x) x \psi(x) dx \\ &= A^2 \int_{-\infty}^{\infty} x e^{-\frac{x^2}{\Delta^2}} dx = \frac{1}{2} A^2 \Delta^2 e^{-\frac{x^2}{\Delta^2}} \Big|_{-\infty}^{\infty} = 0\end{aligned}$$

$$\begin{aligned}\langle x^2 \rangle &= \int_{-\infty}^{\infty} \psi^*(x) x^2 \psi(x) dx \\ &= A^2 \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{\Delta^2}} dx \\ &= \frac{1}{2} A^2 (-\Delta^2) \int_{-\infty}^{\infty} x d(e^{-\frac{x^2}{\Delta^2}}) = -\frac{1}{2} A^2 \Delta^2 (x e^{-\frac{x^2}{\Delta^2}} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-\frac{x^2}{\Delta^2}} dx) \\ &= \frac{1}{2} A^2 \Delta^2 \cdot \sqrt{\pi} \Delta^2 = \frac{1}{2} \sqrt{\pi} A^2 \Delta^4\end{aligned}$$

$$I = \int_{-\infty}^{+\infty} e^{-ax^2} dx$$

$$I^2 = \int_{-\infty}^{+\infty} e^{-ax^2} dx \int_{-\infty}^{+\infty} e^{-ay^2} dy$$

$$= \iint e^{-ar^2} r dr d\theta$$

$$= \frac{1}{2a} \cdot 1 \cdot 2\pi = \frac{\pi}{a} \rightarrow I = \sqrt{\frac{\pi}{a}}$$

$$\rightarrow \Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} = \frac{\Delta^{1/2}}{\sqrt{\pi}} A \Delta^{3/2}$$

$$\langle \hat{p} \rangle = \int_{-\infty}^{+\infty} \psi^*(x) (-i\hbar \nabla) \psi(x) dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{+\infty} e^{-ik_0 x} e^{-\frac{x^2}{2\Delta^2}} \nabla (e^{ik_0 x} e^{-\frac{x^2}{2\Delta^2}}) dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{+\infty} e^{-ik_0 x} e^{-\frac{x^2}{2\Delta^2}} e^{ik_0 x} (ik_0 e^{-\frac{x^2}{2\Delta^2}} - \frac{2x}{2\Delta^2} e^{-\frac{x^2}{2\Delta^2}}) dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{\infty} (ik_0 - \frac{x}{\Delta^2}) e^{-\frac{x^2}{\Delta^2}} dx$$

$$= (-i\hbar) A^2 ik_0 \sqrt{\pi} \Delta = \sqrt{\pi} k_0 A^2 \Delta$$

$$\langle \hat{p}^2 \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{p}^2 \psi(x) dx$$

$$= -\hbar^2 \Delta^2 \int_{-\infty}^{\infty} e^{-ik_0 x} e^{-\frac{x^2}{2\Delta^2}} \Delta^2 (e^{ik_0 x} e^{-\frac{x^2}{2\Delta^2}}) dx$$

$$= -\hbar^2 \Delta^2 \int_{-\infty}^{+\infty} e^{-ik_0 x} e^{-\frac{x^2}{2\Delta^2}} \Delta^2 \left(e^{ik_0 x} e^{-\frac{x^2}{2\Delta^2}} (ik_0 - \frac{2x}{2\Delta^2}) \right) dx$$

$$= -\hbar^2 \Delta^2 \int_{-\infty}^{\infty} e^{-ik_0 x} e^{-\frac{x^2}{2\Delta^2}} e^{ik_0 x} e^{-\frac{x^2}{2\Delta^2}} (ik_0 - \frac{2x}{2\Delta^2})^2 dx$$

$$= -\hbar^2 \Delta^2 \int_{-\infty}^{\infty} e^{-\frac{x^2}{\Delta^2}} (ik_0 - \frac{2x}{2\Delta^2})^2 dx$$

$$= -\hbar^2 \Delta^2 \int_{-\infty}^{+\infty} (-k_0^2 - \frac{2ik_0}{\Delta^2} x + \frac{x^2}{\Delta^2}) e^{-\frac{x^2}{\Delta^2}} dx$$

$$\int_{-\infty}^{+\infty} x^2 e^{-\frac{x^2}{\Delta^2}} dx = \frac{1}{2} \int_{-\infty}^{\infty} x e^{-\frac{x^2}{\Delta^2}} dx =$$

§ 1.5 schrodinger 方程

经典力学: $E = \frac{\vec{p}^2}{2m} + V$, $E \rightarrow i\hbar \frac{\partial}{\partial t}$, $\vec{p} \rightarrow i\hbar \nabla$

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}, t) \right] \psi(\vec{r}, t)$$

1. 几率流、几率守恒. 连续性方程

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) + \nabla \cdot \vec{J}(\vec{r}, t) = 0 \quad \text{其中 } \rho(\vec{r}, t) := \psi^*(\vec{r}, t) \psi(\vec{r}, t)$$

$$\begin{cases} i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}, t) \right] \psi(\vec{r}, t) \\ -i\hbar \frac{\partial}{\partial t} \psi^*(\vec{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}, t) \right] \psi^*(\vec{r}, t) \end{cases} \quad \text{①} \cdot \psi^* \quad \text{②} \cdot \psi$$

$$\begin{aligned} \hookrightarrow i\hbar \frac{\partial}{\partial t} [\psi^*(\vec{r}, t) \psi(\vec{r}, t)] &= \left(-\frac{\hbar^2}{2m} \right) [\psi^*(\vec{r}, t) \nabla^2 \psi(\vec{r}, t) - \psi(\vec{r}, t) \nabla^2 \psi^*(\vec{r}, t)] \\ &= -\frac{\hbar^2}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) \end{aligned}$$

$$\begin{aligned} \Rightarrow J &= \frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*) \\ &= \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi) \end{aligned}$$

$$\text{引入相位 } \psi(\vec{r}, t) = |\rho(\vec{r}, t)| e^{-\frac{i\theta(\vec{r}, t)}{\hbar}}$$

$$\begin{aligned} \hookrightarrow J &= \frac{\hbar}{m} \text{Im} \left[|\rho| e^{\frac{i\theta}{\hbar}} \nabla (|\rho| e^{-\frac{i\theta}{\hbar}}) \right] \\ &= \frac{\hbar}{m} \text{Im} \left[|\rho| e^{\frac{i\theta}{\hbar}} \left(\frac{\nabla \rho}{\rho} + |\rho| \frac{i}{\hbar} \nabla \theta \right) e^{-\frac{i\theta}{\hbar}} \right] \\ &= \frac{\hbar}{m} \text{Im} \left(\rho \frac{i}{\hbar} \nabla \theta \right) \\ &= \frac{\rho}{m} \nabla \theta \end{aligned}$$

2. 定态 schrodinger 方程

分离变量: $\psi(\vec{r}, t) = \psi(\vec{r}) f(t)$

$$\hookrightarrow E_n = \frac{i\hbar \frac{df}{dt}}{f(t)} = \frac{1}{\psi(\vec{r})} \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r})$$

$$\text{解得 } f(t) = e^{-\frac{iE_n t}{\hbar}}$$

能量本征态 $\{\psi_n\}$

$$\phi(\vec{r}, t) = \sum_n \phi_n \psi_n(\vec{r}) e^{-\frac{iE_n t}{\hbar}}$$

$$\text{其中 } \phi_n = \int d^3r \psi_n^*(\vec{r}) \phi(\vec{r}, 0)$$

§ 1.6 一维定态问题

1. 无限深势井

$$V(x) = \begin{cases} 0 & x < -\frac{a}{2} \text{ 或 } x > \frac{a}{2} \\ \infty & -\frac{a}{2} < x < \frac{a}{2} \end{cases}$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \psi = Ae^{ikx} + Be^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

$$\psi(\frac{a}{2}) = \psi(-\frac{a}{2}) = 0 \rightarrow \begin{cases} Ae^{\frac{ika}{2}} + Be^{-\frac{ika}{2}} = 0 \\ Ae^{-\frac{ika}{2}} + Be^{\frac{ika}{2}} = 0 \end{cases}$$

$$\text{非平凡解要求: } \begin{vmatrix} e^{\frac{ika}{2}} & e^{-\frac{ika}{2}} \\ e^{-\frac{ika}{2}} & e^{\frac{ika}{2}} \end{vmatrix} = 0 \Rightarrow \sin ka = 0, \quad ka = n\pi$$

$$\text{此时 } Ae^{\frac{in\pi}{2}} + Be^{-\frac{in\pi}{2}} = e^{\frac{in\pi}{2}} (A + Be^{-in\pi}) = e^{\frac{in\pi}{2}} (A + B) = 0 \rightarrow A = -B$$

$$\psi(x) = A(e^{ikx} - e^{-ikx}) = 2iA \sin(\frac{n\pi}{a}x)$$

$$= 2iA e^{-i\frac{n\pi}{2}} \sin \frac{n\pi}{a} (x + \frac{a}{2}), \quad n = 1, 2, \dots$$

$$\text{归一化条件: } \int_{-\infty}^{\infty} |\psi(x)|^2 dx$$

$$= \int_{-\frac{a}{2}}^{\frac{a}{2}} 4A^2 \sin^2 \frac{n\pi}{a} (x + \frac{a}{2}) dx$$

$$= 4A^2 \cdot \frac{a}{2} = 2A^2 a = 1 \rightarrow A = \sqrt{\frac{1}{2a}}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} (x + \frac{a}{2})$$

$$E_n(x) = \frac{\hbar^2 k^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2m a^2}$$

$$\text{正交性: } \int_{-\frac{a}{2}}^{\frac{a}{2}} \psi_m(x) \psi_n(x) dx = \delta_{mn}$$

* 边界条件:

① 一般 $\psi(x)$, $\frac{d\psi}{dx}$ 连续

② $V(x)$ 在边界发散, $\frac{d\psi}{dx}$ 可不连续.

例: δ 函数势的世界案例

$$V(x) = \alpha \delta(x)$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + \alpha \delta(x) \psi(x) = E \psi(x)$$

$$\frac{d^2 \psi(x)}{dx^2} = \frac{2m}{\hbar^2} [\alpha \delta(x) - E] \psi(x)$$

$$\frac{d\psi(0^+)}{dx} - \frac{d\psi(0^-)}{dx} = \frac{2m}{\hbar^2} \lim_{\eta \rightarrow 0} \int_{-\eta}^{\eta} [\alpha \delta(x) - E] \psi(x) dx \quad \int_{-\infty}^{+\infty} \delta(x) dx = 1$$

$$= \frac{2m\alpha}{\hbar^2} \psi(0)$$

$$\psi(0^-) = \psi(0^+)$$

③ $V(x)$ 在世界不发散, 但有跃变. $\frac{d\psi}{dx}$ 连续.

$$\frac{d^2 \psi(x)}{dx^2} = \frac{2m}{\hbar^2} (V - E) \psi(x)$$

$$\rightarrow \frac{d\psi(x_0+0^+)}{dx} - \frac{d\psi(x_0-0^+)}{dx} = \frac{2m}{\hbar^2} \lim_{\eta \rightarrow 0} \int_{x_0-\eta}^{x_0+\eta} (E - V) \psi dx = 0$$

有限.

2. 自由粒子 $V(x) = 0$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi \rightarrow \psi(x) = A e^{ikx}, \quad k = \sqrt{\frac{2mE}{\hbar^2}} = \frac{p}{\hbar}$$

(1) δ 函数归一化: $\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$

$$\delta[a(x-x_0)] = \frac{1}{|a|} \delta(x-x_0)$$

$$f(x) \delta(x-x_0) = f(x_0) \delta(x-x_0)$$

$$\int_{-\infty}^{+\infty} \psi_k^*(x) \psi_{k'}(x) dx = A^2 \int_{-\infty}^{+\infty} e^{-i(k-k')x} dx = 2\pi A^2 \delta(k-k')$$

$$\text{取 } A = \frac{1}{\sqrt{2\pi}}, \quad \psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}$$

$$\int_{-\infty}^{+\infty} dx \delta(x-x_0) \delta(x-x_1) = \delta(x_0-x_1)$$

$$\int_{-\infty}^{+\infty} \psi_k^*(x) \psi_{k'}(x) dx = \delta(k-k') \quad \text{类似 } C_i a_j = \delta_{ij} \quad \int_{-\infty}^{+\infty} dx \left[\frac{\partial^n}{\partial x^n} \delta(x-x_0) \right] f(x) = (-1)^n f(x_0)$$

$$\frac{\partial}{\partial x} \delta(x-x_0) = \pm \frac{1}{a} \frac{\partial}{\partial x} \delta(x-x_0)$$

$$(x-x_0) \frac{\partial}{\partial x} \delta(x-x_0) = -\delta(x-x_0)$$

(2) 周期归一化: $[-\frac{L}{2}, \frac{L}{2}]$

周期性边界条件: $\psi(-\frac{L}{2}) = \psi(\frac{L}{2})$

$$\hookrightarrow e^{\frac{ipL}{\hbar}} = e^{-\frac{ipL}{\hbar}}, \quad e^{\frac{ipL}{\hbar}} = 1, \quad \frac{pL}{\hbar} = 2n\pi$$

$$p_n = \frac{2\pi\hbar}{L} n, \quad \psi_n(x) = A e^{\frac{ip_n x}{\hbar}} = A e^{2\pi i \frac{n x}{L}}$$

$$\text{正交: } \int_{-\frac{L}{2}}^{\frac{L}{2}} \psi_m^*(x) \psi_n(x) dx = \int_{-\frac{L}{2}}^{\frac{L}{2}} A^2 e^{2\pi i \frac{(n-m)x}{L}} dx$$

$$= A^2 \delta_{mn}$$

$L \rightarrow \infty$ 时, 动量取值为连续值.

3. 有限深方势井

$$V(x) = \begin{cases} -V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & x < -\frac{a}{2}, x > \frac{a}{2} \end{cases}$$

1) 束缚态: $E < V(+\infty) = 0$ 且 $\psi(\pm\infty) \rightarrow 0$

① $x < -\frac{a}{2}$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \psi(x) = \underline{Ae^{-\alpha x}} + Be^{\alpha x}, \text{ 其中 } \alpha = \frac{\sqrt{-2mE}}{\hbar}$$

$$\psi(x) = Be^{\alpha x}$$

$$\psi(-\infty) \rightarrow 0, A=0$$

② $-\frac{a}{2} < x < \frac{a}{2}$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = (E - V_0)\psi \rightarrow \psi(x) = \underline{C(e^{ikx} \pm e^{-ikx})}, \text{ 其中 } k = \frac{\sqrt{2m(E-V_0)}}{\hbar}$$

束缚态有确定的宇称 (对称势)

③ $x > \frac{a}{2}$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \rightarrow \psi(x) = De^{-\alpha x} + \underline{Fe^{\alpha x}} \quad \psi(+\infty) \rightarrow 0, F=0$$

$$\psi(x) = De^{-\alpha x}$$

偶宇称: $B=D, +$

$$\text{解} \begin{cases} Be^{\alpha x} & x < -\frac{a}{2} \\ C(e^{ikx} + e^{-ikx}) & -\frac{a}{2} < x < \frac{a}{2} \\ Be^{-\alpha x} & x > \frac{a}{2} \end{cases} = 2C \cos kx$$

$$\text{边界} \begin{cases} Be^{-\alpha \frac{a}{2}} = 2C \cos(k \frac{a}{2}) \\ 2Be^{-\alpha \frac{a}{2}} = 2Ck \sin(k \frac{a}{2}) \end{cases}$$

B, C 的非平凡解要求:

$$\begin{vmatrix} e^{-\alpha \frac{a}{2}} & 2\cos(\frac{ka}{2}) \\ \alpha e^{-\alpha \frac{a}{2}} & 2k \sin(\frac{ka}{2}) \end{vmatrix} = e^{-\alpha \frac{a}{2}} 2k \sin \frac{ka}{2} - 2\cos \frac{ka}{2} \alpha e^{-\alpha \frac{a}{2}} \\ = 2e^{-\alpha \frac{a}{2}} \cos \frac{ka}{2} [k \tan(\frac{ka}{2}) - \alpha] = 0$$

$$\text{解得 } \alpha = k \tan(\frac{1}{2}ka) \quad (\text{偶宇称})$$

奇宇称: $B = -D, -$

$$\text{解} \begin{cases} B e^{ax} & x < -\frac{a}{2} \\ C(e^{ikx} - e^{-ikx}) = 2C' \sin kx & \\ -B e^{-ax} & x > \frac{a}{2} \end{cases}$$

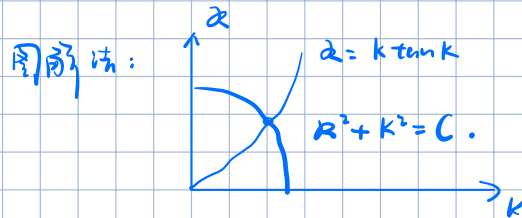
$$\text{边界} \begin{cases} B e^{-k \frac{a}{2}} = 2C' \sin(-\frac{1}{2}ka) \\ a B e^{ax} = 2KC' \cos(-\frac{1}{2}ka) \end{cases}$$

$$\begin{vmatrix} e^{-k \frac{a}{2}} & 2 \sin(-\frac{1}{2}ka) \\ a e^{-k \frac{a}{2}} & 2K \cos(-\frac{1}{2}ka) \end{vmatrix} = e^{-k \frac{a}{2}} \cos(\frac{1}{2}ka) \left[k + a \tan(\frac{1}{2}ka) \right] = 0$$

解得 $a = -K \cot(\frac{1}{2}ka)$ (奇宇称)

$$a = \frac{\sqrt{2m(E-V_0)}}{\hbar}, \quad k = \frac{\sqrt{2m(E-V_0)}}{\hbar}$$

$$\hookrightarrow k^2 + a^2 = -\frac{2mV_0}{\hbar^2}$$



一维束缚态:

(1) 本征能量不简并 假设存在能量相同的态 ψ_1, ψ_2

$$\text{证: } E_1 = E_2 = E, \quad \frac{1}{\psi_1} \frac{d^2 \psi_1}{dx^2} = \frac{2m}{\hbar^2} [V(x) - E] = \frac{1}{\psi_2} \frac{d^2 \psi_2}{dx^2}$$

$$\hookrightarrow \frac{1}{\psi_1} \frac{d\psi_1}{dx} - \frac{1}{\psi_2} \frac{d\psi_2}{dx} = C$$

$$\text{取 } x \rightarrow \infty, \quad \psi_1(\infty) = \psi_2(\infty) = 0, \quad C = 0$$

$$\Rightarrow \frac{1}{\psi_1} \frac{d\psi_1}{dx} = \frac{1}{\psi_2} \frac{d\psi_2}{dx}, \quad \text{解得 } \ln \psi_1 = \ln \psi_2 + C, \quad \psi_1 = \lambda \psi_2$$

由归一化条件可确定 $\lambda = 1$, ψ_1, ψ_2 为同一个态.

(2) 第 n 个束缚态有 $n-1$ 个节点.

引理: 束缚态的 $\psi(x)$ 为实函数. 那 ~ 也可以吗. 只要满足定态. ~ 为实

$$\frac{1}{\psi_1} \frac{\hbar^2}{2m} \frac{d^2 \psi_1}{dx^2} + E_1 = V(x) = \frac{1}{\psi_2} \frac{d^2 \psi_2}{dx^2} + E_2 \quad (E_1 < E_2)$$

$$\hookrightarrow \frac{\hbar^2}{2m} \frac{d}{dx} \left(\psi_1 \frac{d\psi_2}{dx} - \psi_2 \frac{d\psi_1}{dx} \right) = (E_2 - E_1) \psi_1 \psi_2$$

(a) 命题: 每两个连续 $\psi_1(x)$ 节点之间必有一个 $\psi_2(x)$ 的节点.

假设 $x=a, x=b$ 是 $\psi_1(x)$ 的两个连续节点.

$$\int_a^b \text{LHS} = \frac{\hbar^2}{2m} \left[\psi_2(a) \frac{d\psi_1}{dx}(a) - \psi_2(b) \frac{d\psi_1}{dx}(b) \right] = (E_2 - E_1) \int_a^b \psi_1(x) \psi_2(x) dx$$

2) 散射态 $E > V(+\infty) = 0$

① $x < -\frac{a}{2}$, $\psi(x) = Ae^{ikx} + Be^{-ikx}$, $k = \frac{\sqrt{2mE}}{\hbar}$

② $-\frac{a}{2} < x < \frac{a}{2}$, $\psi(x) = Ce^{ikx} + De^{-ikx}$, $k = \frac{\sqrt{2m(E+V_0)}}{\hbar}$

③ $x > \frac{a}{2}$, $\psi(x) = Fe^{ikx}$

边界

$$\begin{cases} Ae^{-ik\frac{a}{2}} + Be^{ik\frac{a}{2}} = Ce^{-ik\frac{a}{2}} + De^{ik\frac{a}{2}} \\ ik(Ae^{-ik\frac{a}{2}} - Be^{ik\frac{a}{2}}) = ik(Ce^{-ik\frac{a}{2}} - De^{ik\frac{a}{2}}) \\ Ce^{ik\frac{a}{2}} + De^{-ik\frac{a}{2}} = Fe^{ik\frac{a}{2}} \\ ik(Ce^{-ik\frac{a}{2}} - De^{ik\frac{a}{2}}) = ikFe^{ik\frac{a}{2}} \end{cases}$$

解得

$$B = i \frac{\sin ka}{2kK} (K^2 - K'^2) F$$

$$F = \frac{e^{-ika} A}{\cos(ka) - i \frac{\sin ka}{2kK} (K^2 + K'^2)}$$

透射系数 $T := |J_t|$

$\psi_t = Fe^{ikx}$, $J_t = \frac{F^2}{m} e^{ikx} \frac{\partial}{\partial x} (-\hbar kx)$

$\psi_i = Ae^{ikx}$, $J_i = \frac{A^2}{m} e^{ikx} \frac{\partial}{\partial x} (-\hbar kx)$

$$T = \left| \frac{F}{A} \right|^2$$

$$= \frac{1}{\cos^2(ka) + \sin^2(ka) \frac{(K^2 + K'^2)^2}{4K^2 K'^2}} = \frac{1}{1 + \sin^2(ka) \frac{(K^2 - K'^2)^2}{4K^2 K'^2}}$$

$$= \frac{4K^2 K'^2}{4K^2 K'^2 + (K^2 - K'^2)^2 \sin^2(ka)} = \frac{\frac{4}{E_0} (2m)^2 (E + V_0) E}{\frac{4}{\hbar^2} (2m)^2 (E + V_0)^2 + (\frac{2m}{\hbar^2} V_0)^2 \sin^2(ka)}$$

$$= \frac{4E(E + V_0)}{4E(E + V_0) + V_0^2 \sin^2(\frac{\sqrt{2m(E+V_0)}}{\hbar} a)}$$

反射系数 $R := |J_r|$

$\psi_r = Be^{-ikx}$, $J_r = \frac{B^2}{m} e^{-ikx} \frac{\partial}{\partial x} (\hbar kx)$

$\psi_i = Ae^{ikx}$, $J_i = \frac{A^2}{m} e^{ikx} \frac{\partial}{\partial x} (-\hbar kx)$

$$= \left| \frac{B}{A} \right|^2$$

$$= \left| \frac{F}{A} \right|^2 \cdot \left| \frac{B}{F} \right|^2 = T \cdot \left| \frac{\sin ka}{2kK} (K^2 - K'^2) \right|^2 = \frac{\sin^2(ka) (K^2 - K'^2)^2}{4K^2 K'^2 + (K^2 - K'^2)^2 \sin^2(ka)}$$

$$= \frac{V_0^2 \sin^2(\frac{\sqrt{2m(E+V_0)}}{\hbar} a)}{4E(E + V_0) + V_0^2 \sin^2(\frac{\sqrt{2m(E+V_0)}}{\hbar} a)}$$

$$T + R = 1$$

共振透射: $\frac{a}{\hbar} \sqrt{2m(E + V_0)} = n\pi$

此时 $R = 0$

4. 势井 $V(x) = \begin{cases} V_0 & -\frac{a}{2} \leq x \leq \frac{a}{2} \\ 0 & \text{其它} \end{cases}$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < -\frac{a}{2} \\ Ce^{\alpha x} + De^{-\alpha x} & -\frac{a}{2} < x < \frac{a}{2} \\ Fe^{ikx} & x > \frac{a}{2} \end{cases} \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad \alpha = \frac{\sqrt{2m(V_0-E)}}{\hbar}$$

边界

$$\begin{cases} Ae^{-ik\frac{a}{2}} + Be^{ik\frac{a}{2}} = Ce^{-\alpha\frac{a}{2}} + De^{\alpha\frac{a}{2}} \\ ik(Ae^{-ik\frac{a}{2}} - Be^{ik\frac{a}{2}}) = \alpha(Ce^{-\alpha\frac{a}{2}} - De^{\alpha\frac{a}{2}}) \\ Fe^{ik\frac{a}{2}} = Ce^{-\alpha\frac{a}{2}} + De^{\alpha\frac{a}{2}} \\ ikFe^{ik\frac{a}{2}} = \alpha(Ce^{\alpha\frac{a}{2}} - De^{-\alpha\frac{a}{2}}) \end{cases}$$

解得

透射系数 T

反射系数 R

3. δ 函数势 $V(x) = \gamma \delta(x)$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} \psi(x) + \gamma \delta(x) \psi(x) = E \psi(x)$$

$x=0$ 处, 二阶导数发散, 一阶导数不连续

$$\lim_{\eta \rightarrow 0^+} \int_{-\eta}^{+\eta} \left(\frac{e}{\eta} \int_{-\eta}^{+\eta} \delta(x) = 1 \right)$$

$$\hookrightarrow \frac{d}{dx} \psi(0^+) - \frac{d}{dx} \psi(0^-) = \frac{2m\gamma}{\hbar^2} \psi(0)$$

(1) 散射态 $E > 0$

$$x \neq 0, \quad \frac{d^2 \psi(x)}{dx^2} + k^2 \psi(x) = 0, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} \\ Ce^{ikx} \end{cases} \quad \text{粒子从左边入射.}$$

$$\text{边界条件: } \begin{cases} A + B = C \\ ikC - ik(A - B) = \frac{2m\gamma}{\hbar^2} C \end{cases} \quad \text{解得 } \begin{cases} B = \frac{m\gamma}{i\hbar^2 k - m\gamma} A \\ C = \frac{i\hbar^2 k}{i\hbar^2 k - m\gamma} A \end{cases}$$

$$T = \left| \frac{j_t}{j_i} \right| = \left| \frac{C}{A} \right|^2 = \frac{(\hbar^2 k)^2}{(\hbar^2 k)^2 + (m\gamma)^2} = \frac{2mE\hbar^2}{2mE\hbar^2 + m^2\gamma^2} = \frac{2\hbar^2 E}{2\hbar^2 E + m\gamma^2}$$

$$T = \left| \frac{j_r}{j_i} \right| = \left| \frac{B}{A} \right|^2 = \frac{(m\gamma)^2}{(\hbar^2 k)^2 + (m\gamma)^2} = \frac{m^2\gamma^2}{2mE\hbar^2 + m^2\gamma^2} = \frac{m\gamma^2}{2\hbar^2 E + m\gamma^2}$$

(2) 束缚态 $E < 0$

$$x \neq 0: \quad \frac{d^2 \psi}{dx^2} - \alpha^2 \psi = 0, \quad \alpha = \frac{\sqrt{-2mE}}{\hbar}$$

$$\text{通解: } \psi(x) = Ae^{\alpha x} + Be^{-\alpha x}$$

$$\text{解为 } \psi(x) = \begin{cases} Ae^{\alpha x} & x < 0 \\ Be^{-\alpha x} & x > 0 \end{cases} \quad \text{束缚态 } \psi(\infty) \rightarrow 0$$

$$\text{边界条件: } \begin{cases} A = B \\ -\alpha A - \alpha B = -\frac{2m\gamma}{\hbar^2} A \end{cases}$$

$$\text{解得 } \alpha = \frac{m\gamma}{\hbar^2}, \quad E = -\frac{m\gamma^2}{2\hbar^2}$$

$$\text{归一化条件: } \int_{-\infty}^{+\infty} dx |\psi(x)|^2 = 2 \int_0^{+\infty} dx A^2 e^{-2\alpha x} = \frac{A^2}{\alpha} = 1$$

$$\hookrightarrow A = \sqrt{\alpha} = \sqrt{\frac{m\gamma}{\hbar^2}}$$

1b 1/2:

1. 1) 1b - 1b 1/2 1/2:

$$\int_{-\infty}^{\infty} |\alpha x e^{-\beta x} e^{i\gamma x/\hbar}|^2 dx = 1$$

$$(2) \langle \hat{x} \rangle = \int dx \langle \psi^* | \hat{x} | \psi \rangle = \int_{-\infty}^{+\infty} \alpha x \cdot x |\psi(x)|^2$$

$$\begin{aligned} \langle \hat{p} \rangle &= \int_{-\infty}^{+\infty} \alpha x \langle \psi^* | \hat{p} | \psi \rangle \\ &= -i\hbar \int_{-\infty}^{+\infty} \alpha x \alpha^2 x e^{-\beta x} \frac{\partial}{\partial x} (x e^{-\beta x}) \end{aligned}$$

$$(3) \frac{d|\psi(x)|^2}{dx} = 0$$

$$(4) \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x) = i\hbar \frac{\partial}{\partial t} \psi(x)$$

$$\begin{aligned} \hookrightarrow V(x) &= \frac{1}{\psi(x)} \left(i\hbar \frac{\partial}{\partial t} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \psi(x) \\ &= i\hbar \cdot \frac{i}{\hbar} + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} (x e^{-\beta x}) \cdot \frac{1}{x e^{-\beta x}} \\ &= -\gamma + \frac{\hbar^2}{2m} \beta \left(\beta - \frac{2}{x} \right) \end{aligned}$$

$$2. 1) \int_{-\infty}^{+\infty} |\psi(x)|^2 dx = \int_0^a |\psi(x)|^2 dx = 1$$

$$\psi(x, 0) = A \sin \frac{\pi x}{a} + \frac{1}{2} A e^{i\varphi} \sin \frac{2\pi x}{a}$$

$$E_n =$$

$$\psi(x, t) = \psi_0 e^{-\frac{iE_1 t}{\hbar}}$$

$$\langle \hat{T} \rangle = \langle \hat{p} \rangle$$

$$p(x) = \langle \psi^* | \hat{p} | \psi \rangle$$

$$T(x) dT = p(x) dp$$

$$(2) \langle \hat{x} \rangle =$$

$$3. \quad (1) \quad \begin{cases} Ae^{ikx} + Be^{-ikx} & , 0 \leq x < a \quad k = \frac{\sqrt{2mE}}{\hbar} \\ Ce^{ikx} & , x \geq a \quad k = \frac{\sqrt{2m(E-V_0)}}{\hbar} \end{cases}$$

边界: $\begin{cases} A + B = C \\ Ae^{ika} + Be^{-ika} = Ce^{ika} \\ k(Ae^{ika} - Be^{-ika}) = kCe^{ika} \end{cases}$

$$\Rightarrow \begin{cases} 2iA \sin ka - Ce^{ika} = 0 \\ 2kA \cos ka - kCe^{ika} = 0 \end{cases} \quad e^{iy} = \cos y + i \sin y$$

非零解: $\begin{vmatrix} 2i \sin ka & -e^{ika} \\ 2k \cos ka & -e^{ika} \end{vmatrix} = 2e^{ika} (k i \sin ka - k \cos ka) = 0$

$$\Rightarrow k = -ik \cot(ka)$$

(2) 上述方程仅有 - 解。

$$4. \quad (1) \quad \psi(x) = \begin{cases} Ae^{kx} + Be^{-kx} & k = \frac{\sqrt{2mE}}{\hbar} \\ Ce^{kx} & k = \frac{\sqrt{2m(E-V_0)}}{\hbar} \end{cases}$$

$$\begin{cases} A + B = C \\ k(A - B) = kC \end{cases} \rightarrow \begin{cases} B = \frac{k-k}{k+k} A \\ C = \frac{2k}{k+k} A \end{cases}$$

$$R = \left| \frac{B}{A} \right|^2, \quad T = \left| \frac{C}{A} \right|^2 \cdot \frac{k}{k}$$

$$(2) \quad \begin{cases} A + B = C \\ \frac{d\psi}{dx} \Big|_{x=0^+} - \frac{d\psi}{dx} \Big|_{x=0^-} = \frac{2m}{\hbar^2} (-i\delta \psi(0)) \end{cases}$$

$$\Rightarrow |kC - k(A-B)| = \frac{2m}{\hbar^2} (-i\delta C)$$

$$\begin{aligned} B &= \frac{k-k-\beta}{k+k+\beta} A, \quad \beta = \frac{2m\delta}{\hbar^2} \\ C &= \frac{2k}{k+k+\beta} A \end{aligned}$$

$$R = \left| \frac{B}{A} \right|^2 = \frac{(k-k-\beta)^2}{(k+k+\beta)^2}$$

$$T = \left| \frac{C}{A} \right|^2 \cdot \frac{k}{k} = \frac{4k^2}{(k+k+\beta)^2} \cdot \frac{k}{k}$$

$$\begin{aligned} A &= 1 - R - T = \frac{1}{(k+k+\beta)^2} \left[(k+k+\beta)^2 - (k-k-\beta)^2 - 4k^2 \right] \\ &= \frac{4k\beta}{(k+k+\beta)^2} \end{aligned}$$

5. (1) 基态, 无节点. $\rightarrow$ 只解偶宇称 $\frac{\partial \psi}{\partial x} \Big|_{x=0} = 0$

$$\psi(x) = \begin{cases} A e^{kx} & x < -a \\ B e^{kx} + B e^{-kx} = 2B \cosh kx & -a < x < a \\ A e^{-kx} & x > a \end{cases}$$

$$k = \frac{\sqrt{2mV_0}}{\hbar}$$

$$\begin{cases} A e^{-ka} = 2B \cosh ka \\ 2kB \sinh kx - kA e^{-ka} = \frac{2m}{\hbar^2} V_0 \psi(a) = \frac{2m}{\hbar^2} V_0 2B \cosh ka \end{cases}$$

$$\begin{cases} A e^{-ka} - 2B \cosh ka = 0 \\ kA e^{-ka} - \left(\frac{2mV_0}{\hbar^2} \cosh ka - 2k \sinh ka \right) B = 0 \end{cases}$$

$$\rightarrow k e^{-ka} \left[2k(\sinh ka + \cosh ka) - \frac{2mV_0}{\hbar^2} \cosh ka \right] = 0$$

解得 $k =$